

Firza Chandra Sandjaya Putra

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Data Engineer, AI Engineer, and ML Engineer with ~4 years of expertise in building end-to-end data pipelines and scalable MLOps frameworks. I focus on the evolution of data infrastructure into intelligent automation, specializing in the deployment and optimization of ML models that accelerate workflows. Proven track record in Telco and Consumer brands of delivering high-impact systems that drive automated, data-led decisions.

WORK EXPERIENCE

PT Hypefast Karya Nusantara – Jakarta, Indonesia

March 2025 - Now

Data Engineer

- **Designed L0–L2 ETL pipelines** ingesting order and customer data from 4 marketplace APIs & Google Sheets to BigQuery for **11+ brands**, with **Flask-based token management** on Cloud Run and PostgreSQL for API authentication, enabling unified analytics and reporting.
- Optimized scalable **product recommendation systems** on Shopee Marketplace (NCF, Replenishment, and Baby Cloth) using TensorFlow/Keras, scikit-learn, and FastAPI on GCP (Vertex AI), **driving approximately IDR 300M/month GMV**.
- Created an end-to-end pipeline for a **smart voucher optimization system** using K-Means clustering and linear programming for budget-constrained allocation, **achieving 200× average monthly ROI per brand**.
- **Automated financial workflows** for 3 brands by integrating Jubelio (OMS) with Jurnal.id via API, **reducing invoice processing from days to <5 minutes** for hundreds of daily invoices, and **accelerating cash flow** through faster AR recognition.
- Engineered an **AI-based OCR** system using GCP Document AI (evaluated against Mistral AI), automating invoice and receipt processing, **reducing extraction time from 8 hours to 5 minutes (96× faster)** with 90%+ accuracy.
- **Delivered Metabase dashboards** for Finance, Data Science, and Business Operations teams to monitor catalog performance, advertising AI, and customer data platform.

PT IDstar Cipta Teknologi (Placement in PT XL Axiata Tbk.) – Jakarta, Indonesia

July 2022 - March 2025

Data Engineer and Analytics / ETL Developer

- Developed an end-to-end telco user behavior & credit scoring platform using a Medallion Architecture (Bronze/Silver/Gold) on Snowflake; built PySpark ETL pipelines on AWS (S3, EC2, Lambda) **processing 1B+ daily records with schema evolution**, partition pruning, and data quality gates enforced at each layer.
- Built a monthly PySpark-based lead generation pipeline to process and **enrich millions of XL Axiata subscriber profiles** for targeted bank product acquisition.
- Designed a dashboard to monitor XL's API Economy performance for external partners, **improving API response times by 30%, success rates by 15%, and driving 10% revenue growth from API transactions**.

Telkom Indonesia – Jakarta, Indonesia

February 2021 - August 2021

Data Scientist Internship

- Support data cleansing, analyzed an exploratory dataset, reported, and interpreted the results to drive insights and data-driven decision-making on T-Money within 1 month.
- Developed a business model for new digital products, KALISA, a waqf blockchain system, and the result was implemented into the new KALISA business model within three months.

EDUCATION

Bandung Institute of Technology (ITB), Bandung

Bachelor's Degree - Industrial Engineering

- **Paid assistant of the laboratory:** Production System Laboratory (LSP) on Faculty of Industrial Technology

Online Course/ Self-Learning Licenses

- **Coursera** - Python for data science and Artificial Intelligence (AI)
- **Rice University** - Data Descriptors, Statistical Distributions, and Application to Business Decisions
- **JLPT** - N3 Japanese Proficiency

PROJECT PORTFOLIO AND SKILLS

Freelance & Projects:

- **Tableau for Data Analysis 101** - Create E-learning Course Contributor at Eduvora
- **Social Media Scraping Integration** - Integrations between 5 platforms: Facebook, X, Instagram, TikTok, and YouTube for Marketing Agency research purposes with ETL using MongoDB.
- **Google News Scraping and Parsing (AI Engineering)** - Automate extraction, integrate GEMINI AI, DIFFBOT, and APIFY for parsing, and aggregate news articles for sentiment and trend analysis for Marketing Agency research purposes.
- **Debt Collection Performance Analytics** - Built an end-to-end pipeline and Looker Studio, identified critical bottlenecks with actionable recommendations
- **Writer Density Index (WDI)** - Quantitative Analyst Developed a PCA-based composite model (68.2% variance explained) to analyze regional writing ecosystem strength across Indonesia, delivering policy-ready insights for strategic decision-making.
- **Strategic Valuation & Growth Outlook of Bilibli Pitch Deck** - Built an end-to-end process delivering strategic analysis, financial forecasting, and DCF & comparable valuation to support investment decision-making and highlight re-rating potential through profitability improvement.
- **Cylinder Lifetime & Rehouse Monitoring Dashboard** - Built a preventive maintenance dashboard for excavator cylinders, tracking lifetime status, rehousing timeline, and stock vs demand analysis to optimize scheduling and reduce downtime.
- **Discourse Network Analysis (DNA) and Social Network Analysis (SNA)** - Built a multi-platform pipeline DNA and SNA with pyvis and Chart.js interactive dashboard, parsing with Gemini LLM to analyze Indonesian nuclear energy policy discourse.
- **Affiliate Commission Performance Review Analysis** - Diagnosed a 34% monthly revenue decline by analyzing conversion funnel leaks and cohort retention collapse (23% to 9%), delivering a strategic recovery roadmap to reclaim Rp51.8M.

Tech Stack and Technical Skills/Competencies:

- **AI/LLM:** LangChain, LangGraph, OpenAI API, Anthropic Claude API, Gemini API, FAISS, RAG, SentenceTransformer, OpenCV
- **ML/Data Science:** TensorFlow/Keras, PyTorch, scikit-learn, pandas, numpy, PySpark
- **Backend:** FastAPI, Flask, SQLAlchemy, REST API
- **On-Premise:** Cloudera (CDH), Apache Airflow, Kafka, Hive, Impala, Hadoop (HDFS, YARN)
- **Cloud:** AWS: EC2, S3, Lambda, ECS/ECR, Snowflake. GCP: Vertex AI, BigQuery, Cloud Run, Cloud Storage, Pub/Sub, Document AI, Composer
- **Databases:** PostgreSQL, MySQL, MongoDB Atlas
- **Infra & DevOps:** Docker, Kubernetes, BitBucket CI/CD, Linux, Git
- **BI & Visualization:** Metabase, Looker Studio, Power BI, Tableau, Library seaborn & matplotlib
- **Tools:** Google Sheets, Apps Script, Jira, Confluence, N8N, dbt, Bitwarden
- **Programming Language:** Python, SQL, JavaScript, Java

LEADERSHIP EXPERIENCE

INFINITY ITB (ITB Dance and Performance Art Community) – ITB

Vice President and Head of the General Secretary Department

- Led setting vision, mission, and OKRs, overseeing all organizational activities, driving budget allocation decisions, and representing the organization in external partnerships. Resulted in achieving 95% program execution success across all divisions.
- Led the General Secretariat Department, managing three divisions: Treasury, Administration, and Secretary Management.

Keluarga Mahasiswa Teknik Industri (MTI ITB) – ITB

Head of Staff of People Management & Measurement

- Performed the Social Mapping project for MTI 2018 students to identify the 10 best group combinations.
- Produced appreciation cards for every member at the end of the management period, with 100% of staff recipients.